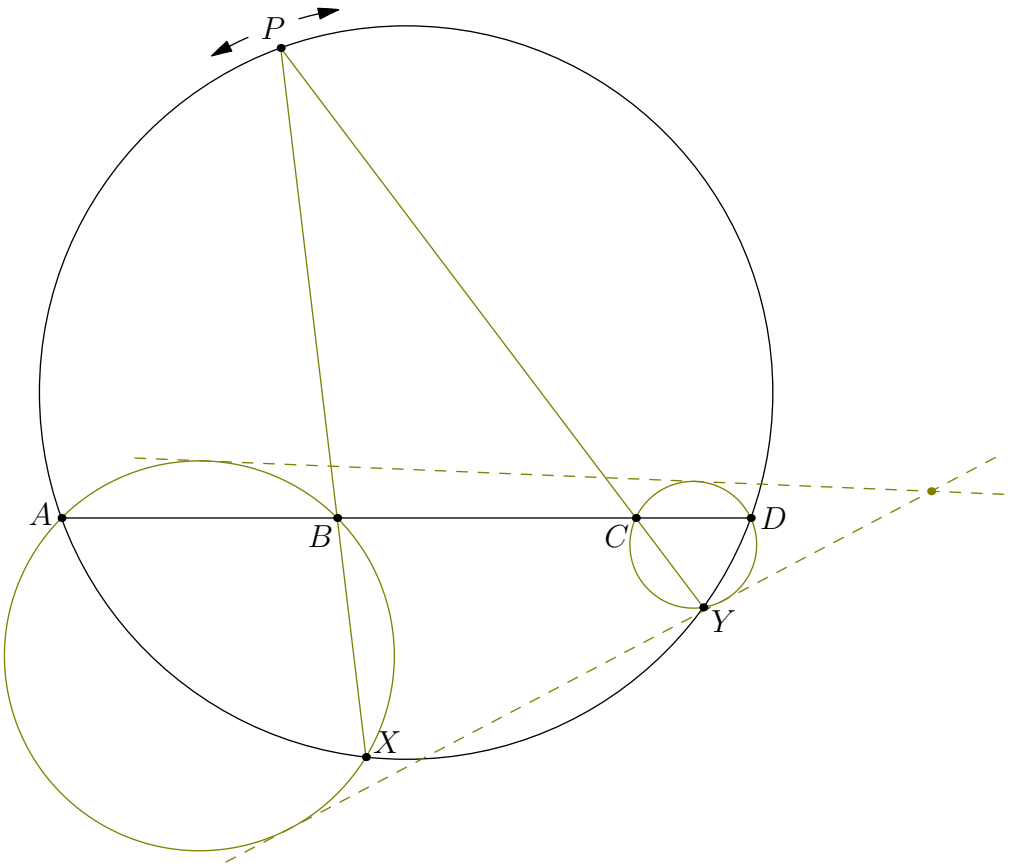
§ Problem Statement

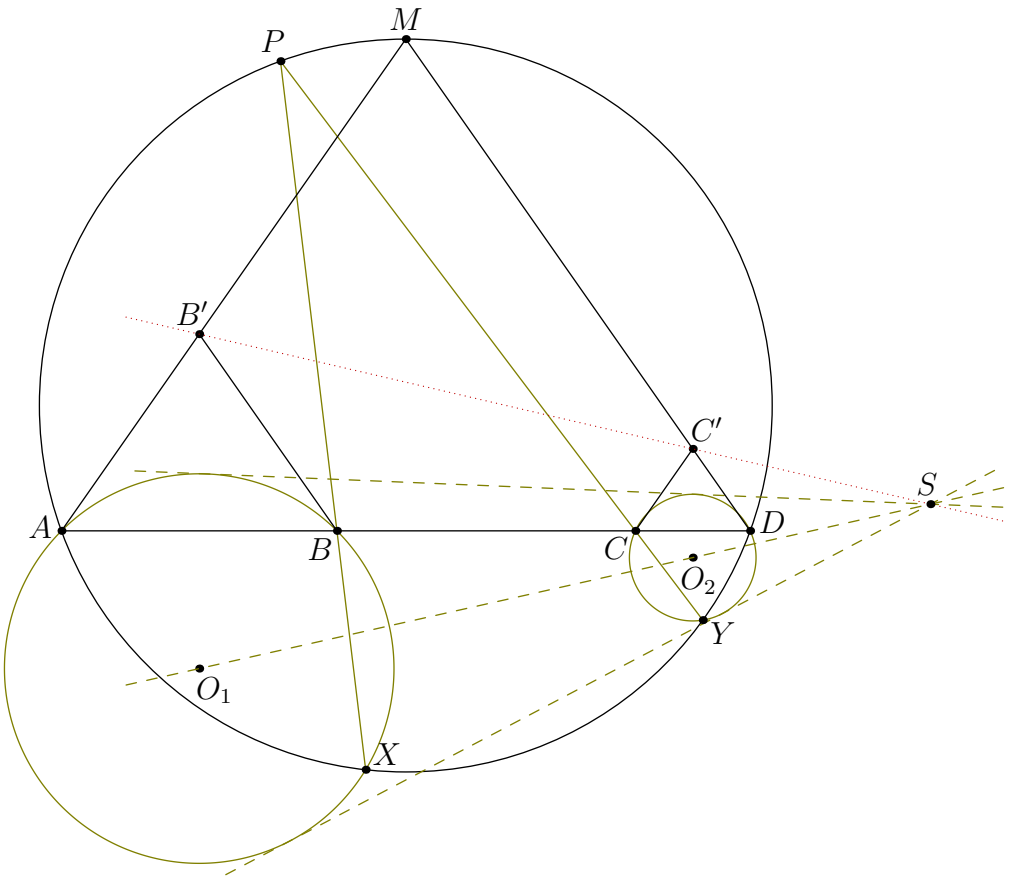
Let B and C be fixed points inside circle ω, and let $\overline{BC}$ meet ω at A and D. Let P be a point on ω, and let $\overline{PB}$ and $\overline{PC}$ meet ω again at X and Y, respectively. Prove that as P varies, (ABX) and (CDY)'s external tangents concur on two fixed lines.

§ Diagram



§ Solution

Let O_1 and O_2 be the centers of (ABX) and (CDY), let r_1 and r_2 be the radii of (ABX) and (CDY), let M be the midpoint of arc AD containing P, let S be the exsimilicenter of (ABX) and (CDY), and let $B' \in \overline{AM}$ and $C' \in \overline{DM}$ satisfy $\overline{BB'} \parallel \overline{DM}$ and $\overline{CC'} \parallel \overline{AM}$. It suffices to show that $S \in \overline{B'C'}$; the other fixed line arises from choosing M to be the other arc midpoint.



By the Law of Sines,

$$\frac{B'O_1}{C'O_2} = \frac{r_1 \cdot \frac{\sin \angle O_1B'B}{\sin B'BO_1}}{r_2 \cdot \frac{\sin \angle O_2C'C}{\sin \angle C'CO_2}} = \frac{r_1}{r_2} \cdot \frac{\sin \angle B'BO_1}{\sin \angle C'CO_2} = \frac{r_1}{r_2},$$

since

$$\begin{aligned} \angle B'BO_1 + \angle C'CO_2 &= \left(180^\circ - \frac{\angle AB'B + \angle AO_1B}{2}\right) + \left(180^\circ - \frac{\angle CC'D + \angle CO_2D}{2}\right) \\ &= 360^\circ - \angle AMD - \angle AXP - \angle DYP \\ &= 180^\circ. \end{aligned}$$

Therefore $B' \cup (ABX)$ and $C' \cup (CDY)$ are homothetic, so their center of homothety, S, must lie on $\overline{B'C'}$.

§ Metadata

This problem was selected as Problem 4 of the 2026 USEMO (Eastern Zone).

- Title: Cyclic Pentagon
- Author: Holden Mui
- Subject: geometry
- Description: the locus of the exsimilicenter of two circles lies on two lines
- Keywords: arc midpoint, exsimilicenter, external tangents, homothety, law of sines, locus
- Difficulty: USAMO 2/5
- Collaborators: Serena An, Ankit Bisain, Maxim Li, Razzi Masroor, Michael Ren, Luke Robitaille, Pitchayut Saengrungkongka, Carl Schildkraut, Issac Zhu
- Date written: December 2020
- Submission history: 2025 TST, 2026 AMM, 2026 USEMO, 2027 USAMO